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A. Appendix: Breakout Model Synthesis (BMS)

Algorithm 3 Breakout Model Synthesis

```
Input,Output: Block B,
Input: Setup restrictions S,
Input: Tile constraints C,
Input: Tile Choice Scheduler TCS
Input: Soften Choice Scheduler SCS
Put B into a fully indeterminate state
Apply setup restrictions S to B
Apply tile constraints C until B is arc consistent
if unable to create initial arc consistent state then
    return failure
end if
Save copy of B to P
while B not fully resolved do
    B' = B
    Choose tile and cell to resolve in B from TCS
    Propagate resolution until B is arc consistent
    if contradiction encountered then
        Revert B back to B'
        Query SCS for a sub-region, R, to soften
        Revert region R in B back to P
        Constraint propagate until B is arc consistent
    end if
    if iteration count has been exceeded then
        return failure
    end if
end while
return success
```